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基本信息

实验室主要从事病毒生物信息学研究，聚焦病毒变异演化和多样性问题，在生物信息学方法开发与大数据分析挖掘方面取得了多项学术成绩，共开发病毒学研究相关的软件和数据库22个，它们均可以免费为科研人员使用；在国际主流学术期刊发表论文70余篇，包括生物信息学权威期刊Bioinformatics (6篇)和Briefings in Bioinformatics (5篇)，病毒学权威期刊Cell Host & Microbe (2篇)、Virologica Sinica (5篇)，生物学权威期刊BMC Biology (1篇)与Science China Life Sciences (1篇)等。

研究成果得到同行的关注和认可，受邀在《传染病信息学概论》中撰写第7章“新型传染病监测预警技术”；担任国家自然科学基金通讯评审和会议评审专家、英国UKRI (UK Research and Innovation) 评审专家；担任Infectious Medicine、Virologica Sinica、Frontiers in Cellular and Infection Microbiology编委以及Briefings in Bioinformatics、Bioinformatics、Genomics Proteomics & Bioinformatics、Molecular Biology and Evolution等50多个权威期刊的审稿人。

先后主持国家自然科学基金项目3项、国家重点研发计划子课题3项、湖南省杰青和优青基金等多个项目；获得2016年湖南省自然科学一等奖（排名第三）与2020年中华医学科技奖二等奖（排名第五）。

教育背景

2007年9月-2012年12月，中国科学院生物物理研究所，生物信息学专业，博士. 导师，蒋太交教授。
2003年9月-2007年7月，山东大学生命科学学院，生物技术专业，学士

工作履历

2013年5月-2016年3月，湖南大学信息科学与工程学院计算机科学系助理研究员；
2016年3月-2017年11月，湖南大学生物学院 助理教授；
2017年12月-2022年12月，湖南大学生物学院 副教授；
2023年1月至今，湖南大学生物学院 教授

本科生课程

承担本科生课程《生物信息学》、《生物信息学综合实践》、《生物信息学实验》、《蛋白质组学》、《生物医学工程导论》

研究领域

本实验室主要从事病毒生物信息学研究，目前主要有两个大的研究方向：

1 传染病的预测预警

包括对于常见传染病如流感病毒与新发突发传染病如新冠病毒和非洲猪瘟病毒等的预测预警。研究问题包括病毒的来源、变异与演化（特别是表型演化）、跨物种感染等，为传染病的预测预警提供方法学与科学依据。该方向主要用到的方法是统计学，机器学习，数学建模等。

2 计算病毒组学

病毒多样性极大，估计有几百万种，但人类对病毒的认识非常少。全球病毒组计划估计有60~80万种病毒可能感染人，但目前已知的感染人的病毒不到1千种。本实验室致力于通过计算生物学的方法研究病毒组的产生与演化机制，病毒组的遗传与功能多样性，病毒与宿主的相互作用，病毒组与人体健康和疾病的关系等等。该方向主要用到组学的方法，主要是多组学数据整合方学，统计学和机器学习方法。

本团队每年招收本科生2~3人，研究生2~3人，博士生1人，博士后多多益善。欢迎有志于生物信息学、大数据分析挖掘、系统生物学、病毒学的同学加入本团队！

科研项目

- 病毒多样性计算分析与预测预警，湖南省杰出青年基金，2024-2026，主持
- 整合多组学数据和机器学习模型鉴定人类病毒组，国家自然科学基金，2024-2027，主持

- 6 反展蛋白的刀云系统地刀机机感病毒基因及酶的组构基轴件刀丁机理，国家自然科学子基金，2014年1月~2017年12月，正分贝贝八
- 8 季节性人流感病毒的疫苗推荐方法研究，湖南大学青年教师成长计划，2013年~2017年，主持
- 9 重大突发动物源性人兽共患病跨种感染与传播机制研究，国家重点研发计划子课题，2016年~2020年，主持
- 10 重要新发突发病原体发生与播散机制研究，国家重点研发计划子课题，2016年~2018年，主持

学术成果

发表论文（#，共同第一；*，通讯作者）：

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- Ping Fu, Zena Cai, Ruina You, Lei Deng, Zhaoyong Li, Zhichao Miao, **Yousong Peng***. Systematic identification and characterization of virus lncRNAs suggests extensive structural mimicry of host lncRNAs. **Briefings in Bioinformatics**, 2025, 26(6): bbaf640
- Wenjie Han, Jinfeng Zeng, Jiayi Shi, Jingru Feng, Yilin Chen, Jialu Zheng, Peiwen Cheng, Ke Zhai, Chi Zhang, Zekai Qiu, Zicheng Cao, Xue Zhang, Jian Chen, Wenping Xie, Xuejie Du, Qing Guo, **Yousong Peng**, Dayan Wang, Yuelong Shu, Taijiao Jiang, Xiangjun Du. Unraveling the mechanism behind the probable extinction of the B/Yamagata lineage of influenza B viruses. **Nature Communications**, 2025, 16(1)

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- Yifan Wu, Zhiyuan Zhang, Xinglian Wang, Xun Liu, Ye Qiu, Xingyi Ge, Zhichao Miao, Xiangxian Meng*, **Yousong Peng***. Virome analysis provides new insights into the pathogenesis mechanism and treatment of SLE disease. **Frontiers in Cellular and Infection Microbiology**, 2024, 14:1484529
- Yifan Wu, **Yousong Peng***. Ten computational challenges in human virome studies. **Virologica Sinica**, 2024
- Ping Fu, Yifan Wu, Zhiyuan Zhang, Ye Qiu, Yirong Wang, **Yousong Peng***. VIGA: an one-stop tool for eukaryotic Virus Identification and Genome Assembly from next-generation-sequencing data. **Briefings in Bioinformatics**, 2024, 25(1), bbaf444 (IF2022=9.5, Q1)
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- Jing-Ying Liao, Yin Shi, Hui Tang, Ping Fu, Jie-Yu Li, Yi-Shuai Zhang, Qian Wu, **You-Song Peng**, Chao-Ting Xiao*. Emerging and characterization of a novel fowl adenovirus 4 strain with open reading frame 19 (ORF19) in diseased chickens from China. **Poultry Science**, 2024, 104702
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- Mi Liu, Jingze Liu, Wenjun Song, **Yousong Peng**, Xiao Ding, Lizong Deng, Taijiao Jiang. Development of PREDAC-H1pdm to model the antigenic evolution of influenza A/(H1N1) pdm09 viruses. **Virologica Sinica**, 2023. (IF 2022=5.5, Q1)
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- Yousong Peng, Aiping Wu, Jing Meng, Lei Yang, Dayan Wang*, Yuelong Shu*, Taijiao Jiang*. Automated recommendation of the seasonal influenza vaccine strain with PREDAC. **Biosafety and Health**, 2020, <https://doi.org/10.1016/j.bsheal.2020.04.002>^(*)^(*)
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开发软件与数据库

软件/数据库	本地/在线版	功能
FluPhenotype	在线	基于基因或蛋白序列，一站式地评估流感病毒的抗原、宿主、致病性和耐药性等特征 (http://www.computationalbiology.cn:18888/IVEW)
PREDAC	在线	包含PREDAC-H3、PREDAC-H1、PREDAC-H5、PREDAC-FluA、PREDAC-B等5个在线软件，分别对应于H3、H1、H5、A型以及B型流感病毒的抗原变异预测与监测平台 (http://www.computationalbiology.cn/home/index.html)
MGFC	在线	非洲猪瘟病毒多基因家族蛋白的分类与功能研究 (http://www.computationalbiology.cn/MGF/home.html)
HumanVirusFinder	本地	基于contig/read序列，预测病毒是否感染人 (https://github.com/Fzhang1992/human-infecting_virus_finder)
PHP	本地与在线版	给定噬菌体的基因组序列预测其宿主 (http://www.computationalbiology.cn/phageHostPredictor/home.html)
cooccurNet	本地	从核酸/蛋白/SNP等数据构建共出现网络的R包 (https://rdrr.io/cran/cooccurNet/)
EVIHVR	在线	人类病毒受体表达、遗传变异和计算鉴定的计算平台 (http://www.computationalbiology.cn/EVIHVR/#/)
3CLP	在线	预测冠状病毒3CL蛋白酶切割位点 (http://www.computationalbiology.cn/3CLPHost/home.html)
vsRNA finder	本地	从小RNA测序数据鉴定高可信度的病毒小RNA分子 (https://github.com/ZenaCai/vsRNAfinder)
ViralReceptor	在线	最全面的哺乳动物病毒蛋白受体数据库 (http://www.computationalbiology.cn:5000/viralReceptor)
phageReceptor	在线	最全面的噬菌体受体数据库 (http://www.computationalbiology.cn/phageReceptor/index.html)
VirusCircBase	在线	首个病毒环状RNA数据库 (http://www.computationalbiology.cn/VirusCircBase/home.html)
HumanVirusDatabase	在线	最全面的人体病毒数据库 (http://computationalbiology.cn/humanVirusBase/#/)
PlantVirusDatabase	在线	最全面的植物病毒组数据库 (http://47.90.94.155/PlantVirusBase/#/)
vsRNAdb	在线	最全面的病毒小RNA数据库 (http://www.computationalbiology.cn/vsRNAdb/)

奖励与荣誉

- 1 2020年中华医学会医学科学技术奖二等奖。蒋太交、舒跃龙、吴爱平、杨磊、**彭友松**、杜向军、王大燕、程根宏、金梅林、秦晓峰。流感防控的关键生物信息技术及其创新应用体系。
- 2 湖南省自然科学一等奖（2016年）。郭新红，谭钟扬，**彭友松**，舒小宝。基于基因组序列数据的分子遗传进化机制及方法。
- 3 2018年湖南省普通高校教师课堂教学竞赛三等奖
- 4 湖南省优秀博士毕业论文指导教师，2023年
- 5 湖南大学优秀博士毕业论文指导教师，2025年

学术兼职

- 1 担任国家自然科学基金通讯评审和会议评审专家、英国UKRI（UK Research and Innovation）评审专家、匈牙利国家科学研究基金会(OTKA) 的评审专家
- 2 Infectious Medicine编委、Virologica Sinica编委、Frontiers in Cellular and Infection Microbiology的Associate Editor
- 3 Briefings in Bioinformatics、Bioinformatics、Genomics Proteomics & Bioinformatics、Molecular Biology and Evolution等40多个权威期刊的审稿人
- 4 湖南省生物信息学会副理事长、病原微生物专委会主委
- 5 中国生物信息学会（筹）重大疾病组学信息学专委会委员、中国微生物学会智能计算微生物专委会委员、中国预防医学会流感预防控制工作委员会委员

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待补充.....